CHINARSOFT · MACHINE VISION

LEGO Vision Inspection

How assembly lines catch missing parts with a camera, in 40 ms, with no AI training

OK NG



THE PROBLEM

A missing part costs far more than the part

Escapes

A defect that reaches the customer means returns, recalls and lost trust.

Line stops

Found late, one wrong part can stop the next station and the whole line.

Tired eyes

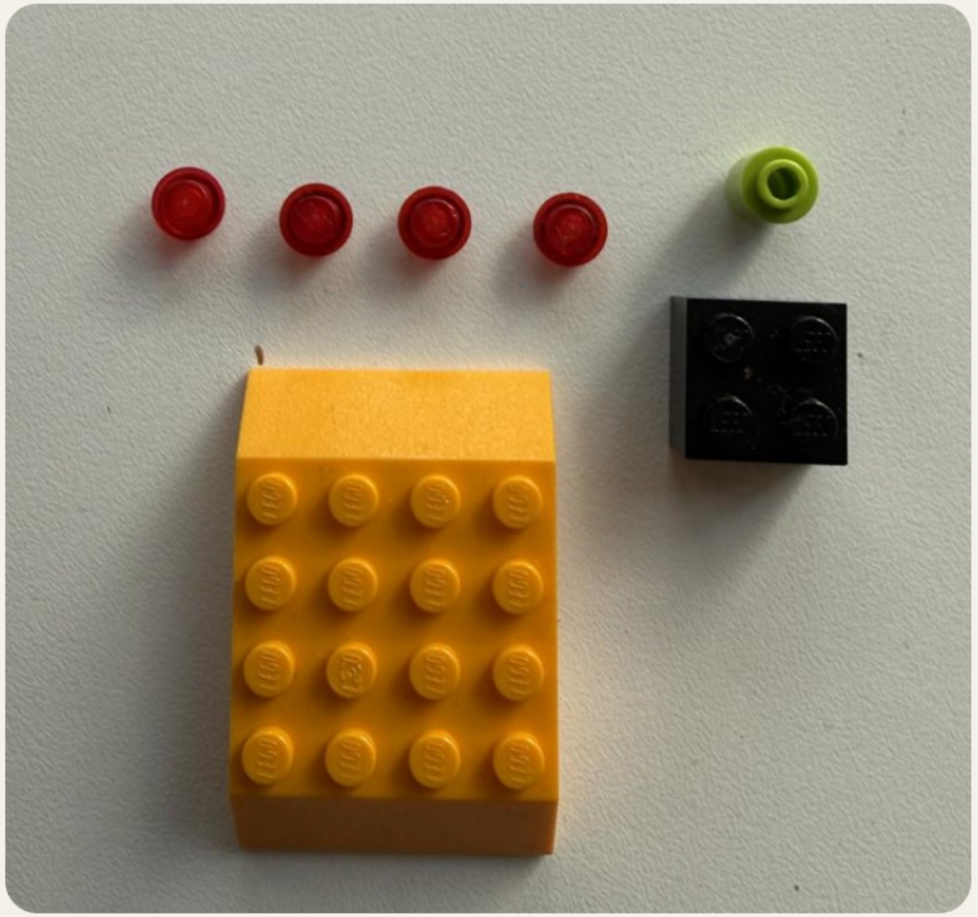
Human inspectors get tired by the end of a shift. A camera does not.

THE PRODUCT

Seven parts, one simple rule

Part	Qty
Orange slope base 4×4	1
Trans-red round 1×1	4
Black brick 2×2	1
Lime round 1×1	1

Red on all four corners. Black in the center. Lime on top of the black brick.



THE LINE

Two cameras guard the assembly

Kit tray
Loose parts

→

Station 1
Kit check: all parts there?

→

Assembly
Operator builds

→

Station 2
Final check: all in place?

OK → the part moves on to packing

NG → rejected, with the reason on screen

THE SETUP

Lighting does most of the work

Camera straight down

HD webcam, fixed, no perspective tricks

Matte white background

Everything that is not white is a part

Light from both sides

About 45°, so no hard shadows

Fixed exposure

Lock auto exposure and white balance

HOW IT WORKS

Five steps from pixels to a decision

01

Segment

Separate parts from the white background

02

Classify

Name each color with HSV ranges

03

Measure

Count the parts or check each position

04

Compare

Against the BOM or the expected layout

05

Decide

OK, or NG with the reason

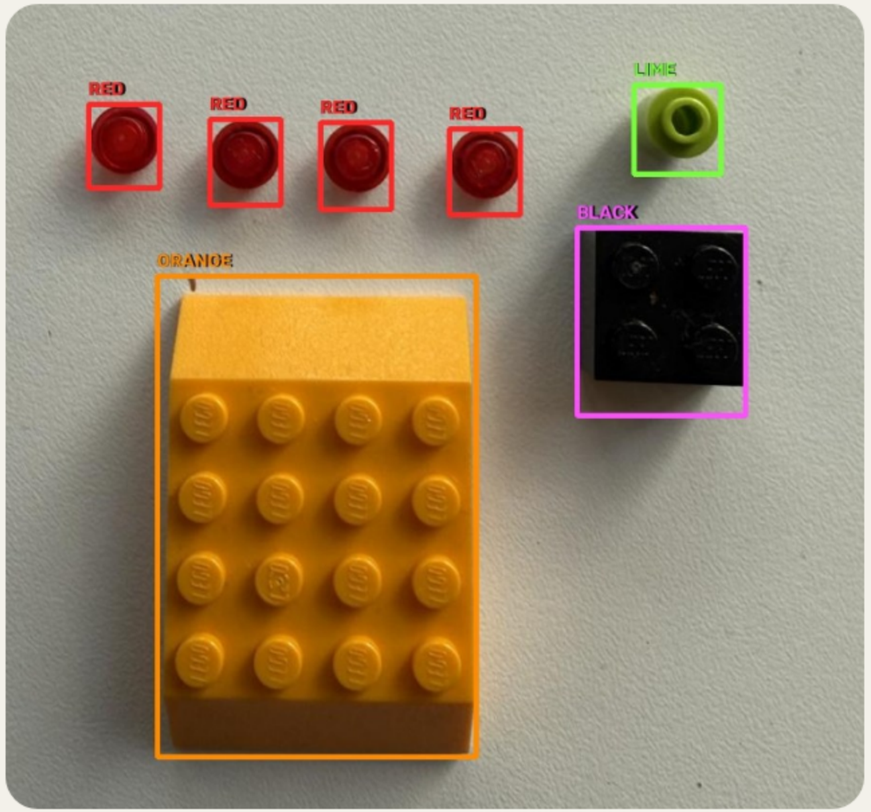
No training data. No AI model. Python + OpenCV, the same idea as industrial vision sensors.

STATION 1 · KIT CHECK

Count by color, compare with the BOM

Color	Found	Expected	Result
Orange	1	1	OK
Red	3	4	Missing 1
Black	1	1	OK
Lime	1	1	OK
Unknown	1	0	Foreign part

Parts must not touch, or two parts count as one. Real kitting trays have pockets for exactly this reason.

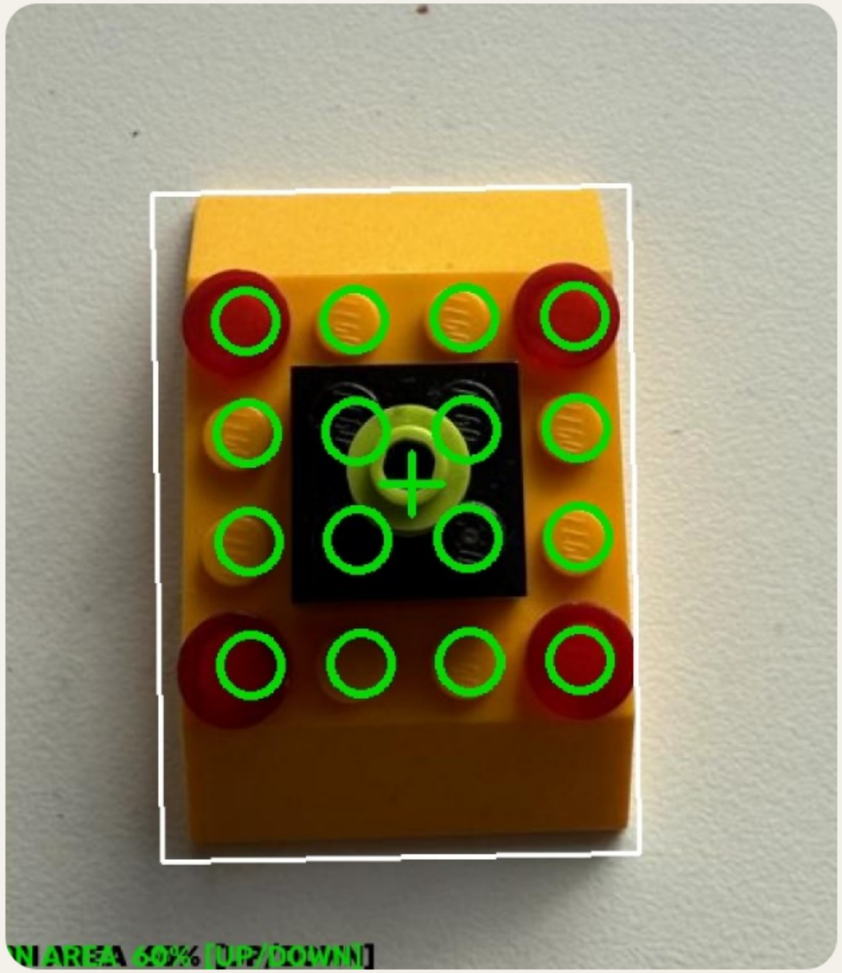


STATION 2 · FINAL CHECK

Check 16 studs and one pin

1. Find the base: the largest object in the box
2. Straighten it with a perspective transform
3. Check the color in 16 small windows, one per stud
4. Check the lime pin in the center
5. Try 0° and 180°, keep the better match

Counting is not enough here: a part can be present but in the wrong place.



WHAT IT CATCHES

Every NG says exactly what is wrong

NG · RED MISSING at BOTTOM-RIGHT

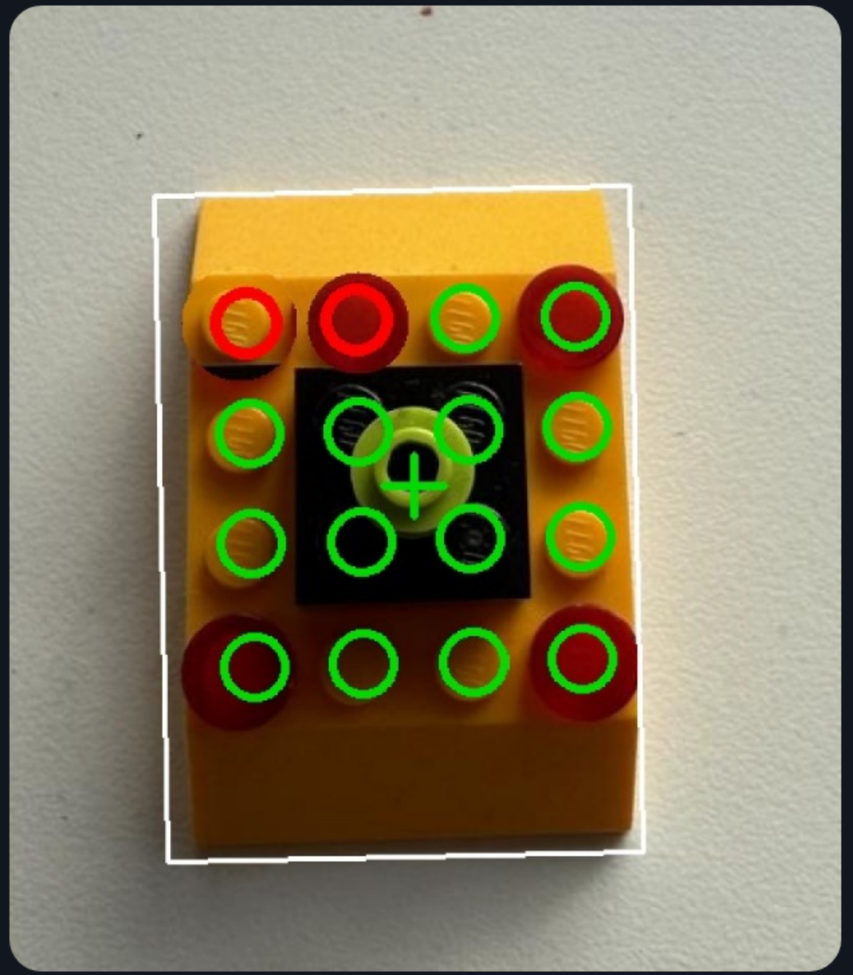
NG · RED MISPLACED at R1C2

NG · BLACK BRICK MISSING / SHIFTED

NG · LIME PIN MISSING

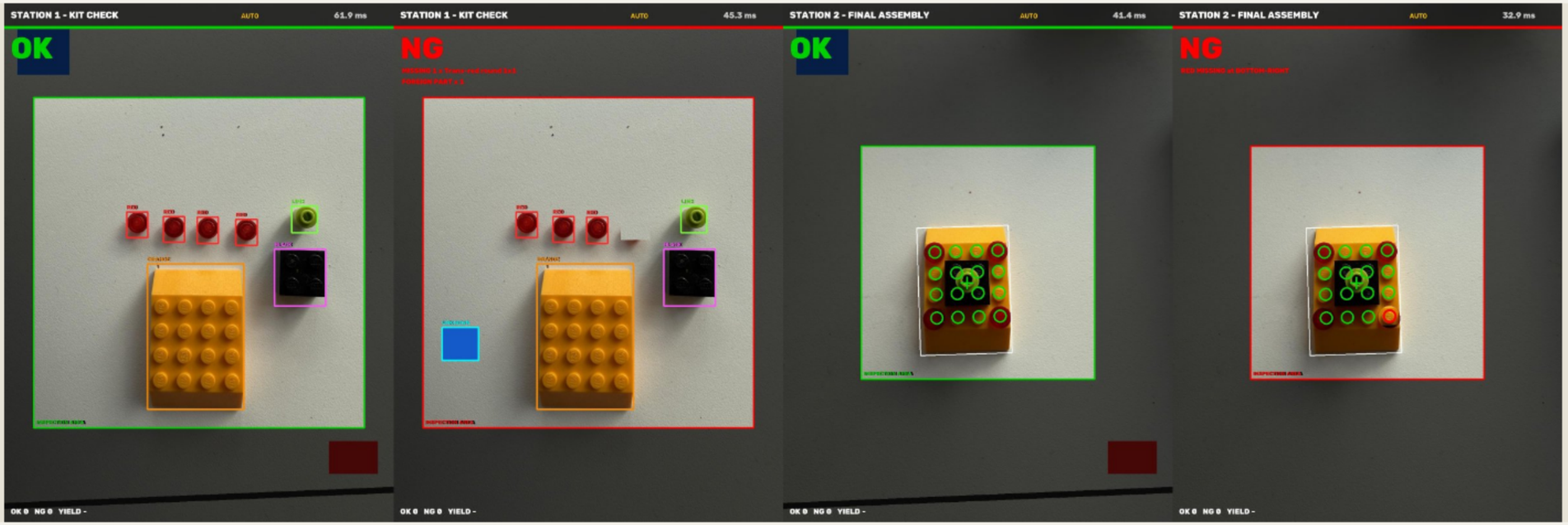
OK · rotated 180°, still correct

Typical cycle time: about 40 ms per part.



INSPECTION AREA

Look only where the part should be



Outside the box is dimmed and ignored, and anything touching its edge is skipped. Resize it live with ↑ and ↓.

NOW LIVE

Let’s break it on purpose

Station 1

- Full kit → OK
- Remove one red → NG
- Add a foreign part → NG

Station 2

- Correct part → OK
- Corner red missing → NG
- Red on the wrong stud → NG
- Black brick shifted → NG
- Lime pin missing → NG
- Rotated 180° → OK

LIMITS & NEXT STEPS

Classic vision is fast, but it trusts color and light

	Classic vision	Trained YOLO	AI vision model
Training data	None	Hundreds of photos	None
Speed	About 40 ms	[measure on Jetson]	[measure]
Sensitive to light	High	Lower	Lower
Explains the NG	Yes, by rule	By class	In words
Episode	This one	Next	After that

Same LEGO, same camera, three methods: a fair comparison in the coming episodes.

BUILD IT YOURSELF

The full code is free on GitHub

`github.com/tahsin-chinarsoft/lego-vision-inspection`

Python + OpenCV · MIT license Sample photos included: no camera needed to try it

Next: the same line with YOLO on a Jetson. Subscribe so you don’t miss it.